Cheat sheet

LLM APIs

Messages, streaming, tools, structured output and reliability

Concepts

Roles	system (rules), user (input), assistant (model turns), tool (results)
Stateless	Resend history each call
Tokens	Billed input + output; check `usage` in responses
Streaming	Server-sent events; show tokens as they arrive
Tool calling	Model returns tool name + JSON args; you execute and return result
Structured output	JSON schema mode; validate with Pydantic
Vision / files	Send images or PDFs as content blocks
Embeddings endpoint	Text → vector for search

Free ways to practise

Google AI Studio	Free, rate-limited Gemini API key; free-tier data may be used to improve Google products
Ollama	Run open models locally, OpenAI-compatible endpoint
Hugging Face	Free models, Spaces and limited free inference
Transformers.js	Run small models in the browser (used in this app's RAG Lab)

Reliability

Errors	400 bad request, 401 auth, 429 rate limit, 5xx server
Retries	Exponential backoff with jitter; idempotent requests
Timeouts	Always set; fall back to a smaller model
Secrets	Environment variables / secret managers; never in client-side code for production

Cost control	Smaller models, caching, max tokens, batch APIs, prompt compression